ANNEX 2

Determination of the required distance for propagation mode (2)

1 Overview

The algorithm given below allows propagation mode (2) path loss, $L_r(p)$ (dB), to be obtained as a monotonic function of rainfall rate, $R(p)$ (mm/h), and with the hydrometeor scatter distance, r_i (km), as a parameter. The model is valid for average annual time percentage (p) in the range 0.001% to 10%. The procedure to determine the hydrometeor scatter contour is as follows:

a) The value of $R(p)$, is determined for the appropriate rain climatic Zones A to Q.

b) Values of $L_r(p)$, are then calculated for incremental values of r_i, starting at the minimum coordination distance d_{min}, in steps of s (km), as described in § 1.3 of the main body of this Appendix. The correct value of r_i is that for which the corresponding value of $L_r(p)$ equals or exceeds the propagation mode (2) minimum required loss $L(p)$. This value of r_i is the propagation mode (2) required distance and is denoted d_r.

c) If the iterative calculation results in r_i equalling or exceeding the appropriate maximum calculation distance (d_{max2}) given in § 2, then the calculation is terminated and d_r is assumed to be equal to d_{max2}. Hence the iteration stops when either of the following expressions becomes true:

$$L_r(p) \geq L(p) \tag{62a}$$

or:

$$r_i \geq d_{max2} \tag{62b}$$

d) The contour for propagation mode (2) is a circle of radius d_r (km) centred on a point along the azimuth of the earth station antenna main beam at a horizontal distance of Δd (km) from the earth station.

2 Maximum calculation distance

As discussed in § 1.5.3 of the main body of this Appendix, it is necessary to set upper limits to the maximum distance used in the iterative calculation of the required distance. The maximum calculation distance to be used for propagation mode (2) (d_{max2}) is latitude dependent and is given in the following equation:

$$d_{max2} = \sqrt{17\,000(h_R + 3)} \quad \text{km}$$

where h_R is defined in equations (74) and (75).

3 Calculation of the propagation mode (2) contour

Determine $R(p)$, the rainfall rate (mm/h) exceeded on average for p% of a year. The world has been divided into a number of rain climatic zones (see Figs. 2, 3 and 4) which show different precipitation characteristics.

FIGURE 2

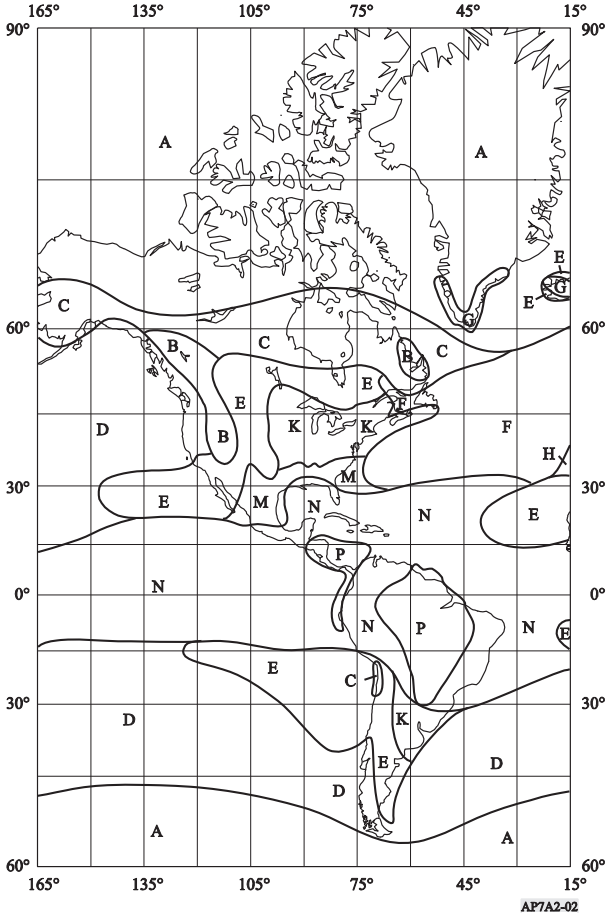


FIGURE 3

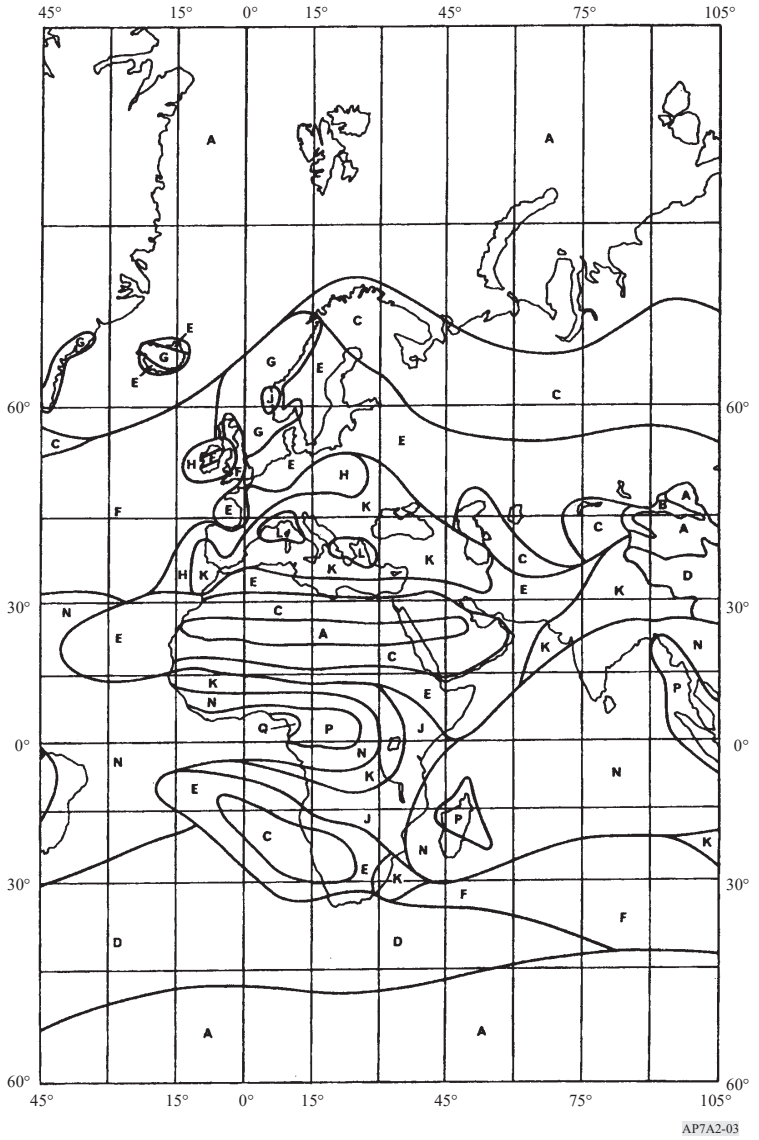
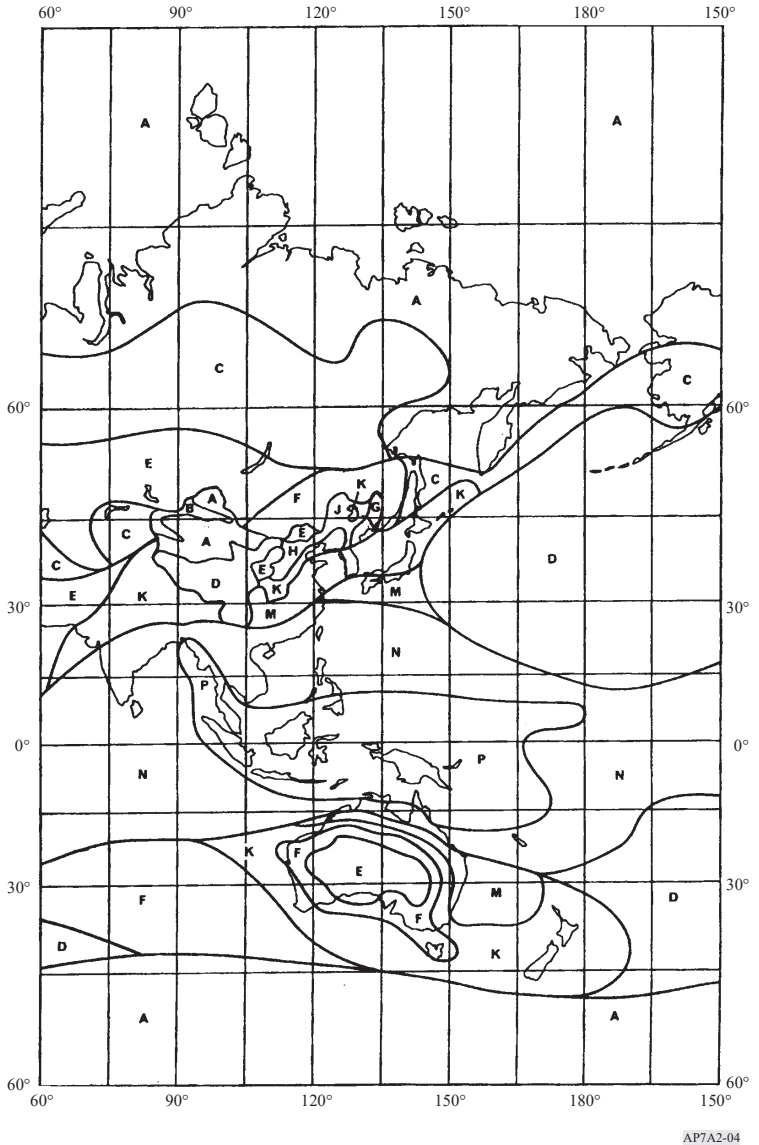


FIGURE 4



The curves shown in Fig. 5 represent consolidated rainfall-rate distributions, each applicable to several of these rain climatic zones.

Determine which rain climatic zone is applicable to the location of the earth station:

– For $0.001\% < p < 0.3\%$ and the applicable rain climatic zone:

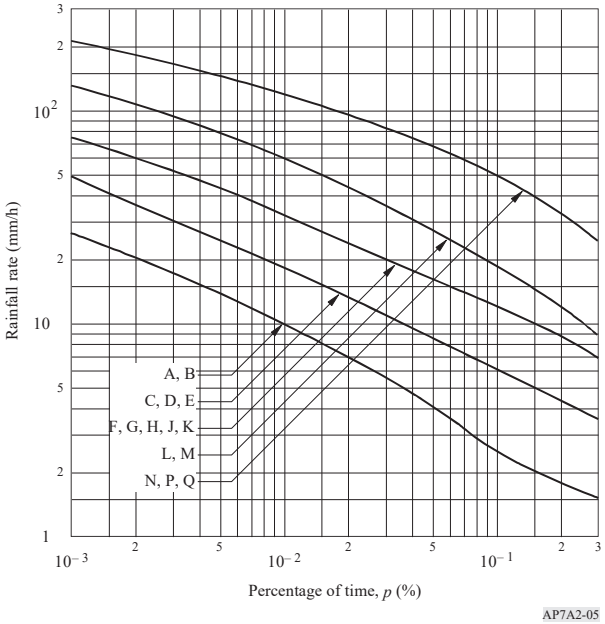
  Determine $R(p)$ either from Fig. 5 or from equations (63) to (67).

– For $p \geq 0.3\%$:

  Use equation (68) with values of $R(0.3\%)$ and p_c obtained from Table 4.

FIGURE 5

Consolidated cumulative distributions of rainfall rate for the rain climatic zones shown in Figs. 2, 3 and 4



Rain climatic Zones A, B

$$R(p) = 1.1\, p^{-0.465} + 0.25\left[\log\left(p/0.001\right)\log^3\left(0.3/p\right)\right] - \left[\left|\log\left(p/0.1\right)\right| + 1.1\right]^{-2} \quad \text{mm/h} \qquad (63)$$

Rain climatic Zones C, D, E

$$R(p) = 2\, p^{-0.466} + 0.5\left[\log\left(p/0.001\right)\log^3\left(0.3/p\right)\right] \quad \text{mm/h} \qquad (64)$$

Rain climatic Zones F, G, H, J, K

$$R(p) = 4.17\, p^{-0.418} + 1.6\left[\log\left(p/0.001\right)\log^3\left(0.3/p\right)\right] \quad \text{mm/h} \qquad (65)$$

Rain climatic Zones L, M

$$R(p) = 4.9\, p^{-0.48} + 6.5\left[\log\left(p/0.001\right)\log^2\left(0.3/p\right)\right] \quad \text{mm/h} \qquad (66)$$

Rain climatic Zones N, P, Q

$$R(p) = 15.6\left(p^{-0.383} + \left[\log\left(p/0.001\right)\log^{1.5}\left(0.3/p\right)\right]\right) \quad \text{mm/h} \qquad (67)$$

TABLE 4

Values of R and p_c for the different rain climatic zones

Rain climatic zone	R (0.3%) (mm/h)	p_c (%)
A, B	1.5	2
C, D, E	3.5	3
F, G, H, J, K	7.0	5
L, M	9.0	7.5
N, P, Q	25.0	10

where:

p_c (%): reference time percentage above which the rainfall rate $R(p)$ can be assumed to be zero.

$$R(p) = R(0.3\%)\left[\frac{\log(p_c/p)}{\log(p_c/0.3)}\right]^2 \qquad (68)$$

Determine the specific attenuation (dB/km) due to rain using values of k and α from Table 5 in equation (70). Values of k and α at frequencies other than those in Table 5 can be obtained by interpolation using a logarithmic scale for frequency, a logarithmic scale for k and a linear scale for α.

TABLE 5

Values of *k* and α for vertical polarization as a function of the frequency

Frequency (GHz)	*k*	α
1	0.0000352	0.880
4	0.000591	1.075
6	0.00155	1.265
8	0.00395	1.31
10	0.00887	1.264
12	0.0168	1.20
14	0.029	1.15
18	0.055	1.09
20	0.0691	1.065
22.4	0.090	1.05
25	0.113	1.03
28	0.150	1.01
30	0.167	1.00
35	0.233	0.963
40	0.310	0.929
40.5	0.318	0.926

Let:

$$R = R(p) \tag{69}$$

Then the specific attenuation (dB/km) due to rain is given by:

$$\gamma_R = k\ R^{\alpha} \tag{70}$$

Calculate the effective diameter of the rain cell:

$$d_s = 3.5\ R^{-0.08} \tag{71}$$

Then, calculate the effective scatter transfer function:

$$R_{cv} = \frac{2.17}{\gamma_R\ d_s}\left(1 - 10^{\frac{-\gamma_R\ d_s}{5}}\right) \tag{72}$$

Calculate the additional attenuation outside the common volume:

$$\Gamma_2 = 631\,k\ R^{(\alpha - 0.5)} \times 10^{-(R+1)^{0.19}} \tag{73}$$

Determine the rain height above ground, h_R (km):

For North America and Europe west of 60° E longitude:

$$h_R = 3.2 - 0.075\,(\zeta - 35) \qquad \text{for} \quad 35 \le \zeta \le 70 \tag{74}$$

where:

ζ: latitude of the coordinating earth station.

For all other areas of the world:

$$h_R = \begin{cases} 5 - 0.075\,(\zeta - 23) & \text{for} \quad \zeta > 23 \quad \text{Northern hemisphere} & (75a) \\ 5 & \text{for} \quad 0 \le \zeta \le 23 \quad \text{Northern hemisphere} & (75b) \\ 5 & \text{for} \quad 0 \ge \zeta \ge -21 \quad \text{Southern hemisphere} & (75c) \\ 5 + 0.1\,(\zeta + 21) & \text{for} \quad -71 \le \zeta < -21 \quad \text{Southern hemisphere} & (75d) \\ 0 & \text{for} \quad \zeta < -71 \quad \text{Southern hemisphere} & (75e) \end{cases}$$

Determine the specific attenuation due to water vapour absorption (a water vapour density of 7.5 g/m^3 is used):

$$\gamma_{wr} = \left[0.06575 + \frac{3.6}{(f - 22.2)^2 + 8.5} \right] f^2\, 7.5 \times 10^{-4} \tag{76}$$

3.1 Iterative calculations

Evaluate equations (77) to (82) inclusive for increasing values of r_i, where r_i is the current distance considered (km) between the region of maximum scattering and the possible location of a terrestrial station and i = 0, 1, 2, ... Continue this process until either of the conditions given in equations (62a) and (62b) is true. Then the rain-scatter required distance d_r is the current value of r_i.

$$r_i = d_{min} + i \cdot s \tag{77}$$

Determine the loss above the rain height, L_{ar} (dB), applicable to scatter coupling:

$$L_{ar} = \begin{cases} 6.5 \left[6\,(r_i - 50)^2 \times 10^{-5} - h_R \right] & \text{for} \quad 6\,(r_i - 50)^2 \times 10^{-5} > h_R & (78a) \\ 0 & \text{for} \quad 6\,(r_i - 50)^2 \times 10^{-5} \le h_R & (78b) \end{cases}$$

Calculate the additional attenuation for the departure from Rayleigh scattering:

$$A_b = \begin{cases} 0.005\,(f - 10)^{1.7} R^{0.4} & \text{for } 10 \text{ GHz} < f < 40.5 \text{ GHz} & (79a) \\ 0 & \text{for } f < 10 \text{ GHz or when } L_{ar} \ne 0 & (79b) \end{cases}$$

Calculate the effective path length for oxygen absorption:

$$d_O = \begin{cases} 0.7\, r_i + 32 & \text{for} \quad r_i < 340 \text{ km} \quad (80a) \\ 270 & \text{for} \quad r_i \geq 340 \text{ km} \quad (80b) \end{cases}$$

Calculate the effective path length for water vapour absorption:

$$d_v = \begin{cases} 0.7\, r_i + 32 & \text{for} \quad r_i < 240 \text{ km} \quad (81a) \\ 200 & \text{for} \quad r_i \geq 240 \text{ km} \quad (81b) \end{cases}$$

Determine the propagation mode (2) path loss, L_r (dB):

$$\begin{aligned} L_r = \; & 168 + 20 \log r_i - 20 \log f - 13.2 \log R - G_x + \\ & A_b - 10 \log R_{cv} + \Gamma_2 + L_{ar} + \gamma_o\, d_o + \gamma_{wr}\, d_v \end{aligned} \quad (82)$$

where:

γ_o: as given in equation (33)

G_x: terrestrial network antenna gain in Tables 7 or 8.

4 Construction of the propagation mode (2) contour

In order to determine the centre of the circular propagation mode (2) contour, it is necessary to calculate the horizontal distance to this point from the earth station, along the azimuth of the earth station antenna main beam axis. The distance, Δd (km), to the centre of the propagation mode (2) contour is given by:

$$\Delta d = \frac{h_R}{2 \tan \varepsilon_s} \quad (83)$$

where:

ε_s: earth station antenna main beam axis elevation angle

and

Δd: shall be limited to the distance (d_r – 50) km.

The propagation mode (2) required distance, d_r, must lie within the range between the minimum coordination distance, d_{min}, and the propagation mode (2) maximum calculation distance, d_{max2}.

Draw the propagation mode (2) contour as a circle of radius d_r (km) around the centre determined above. The propagation mode (2) contour is the locus of points on this circle. However, if any part of the propagation mode (2) contour falls within the contour defined by the minimum coordination distance, this arc of the propagation mode (2) contour is taken to be identical to the contour based on the minimum coordination distance and the propagation mode (2) contour is then no longer circular.